

Window Material for Proximity and Reflective Sensors

Proximity and reflective sensors typically need to be protected by a window. Because these sensors emit and receive near-infrared light, the window material needs to be as transmissive as possible. Vishay recommends the use of window material made from polycarbonate, which is extremely robust, lightweight, available in near-infrared transparent, and impact resistant. Examples of commercially available infrared transmissive polycarbonates include materials sold under the trade names Lexan™, Arla™, and Makrolon®, which are manufactured by various third-party suppliers.

Vishay is not affiliated with any of these suppliers, and does not make any recommendation, warranty or representations for or about any of these suppliers or their products, including but not limited to the feasibility, function, quality, compatibility, availability, compliance, safety etc.. Their inclusion here is for informational purposes only.

A graph of near-infrared transparent polycarbonates with 90 % transmissivity is shown below.

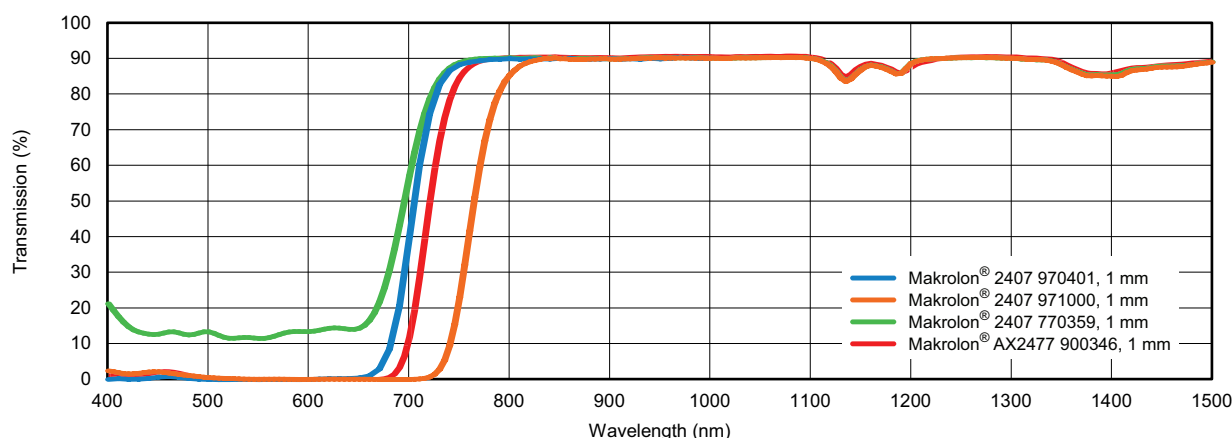


Fig. 1

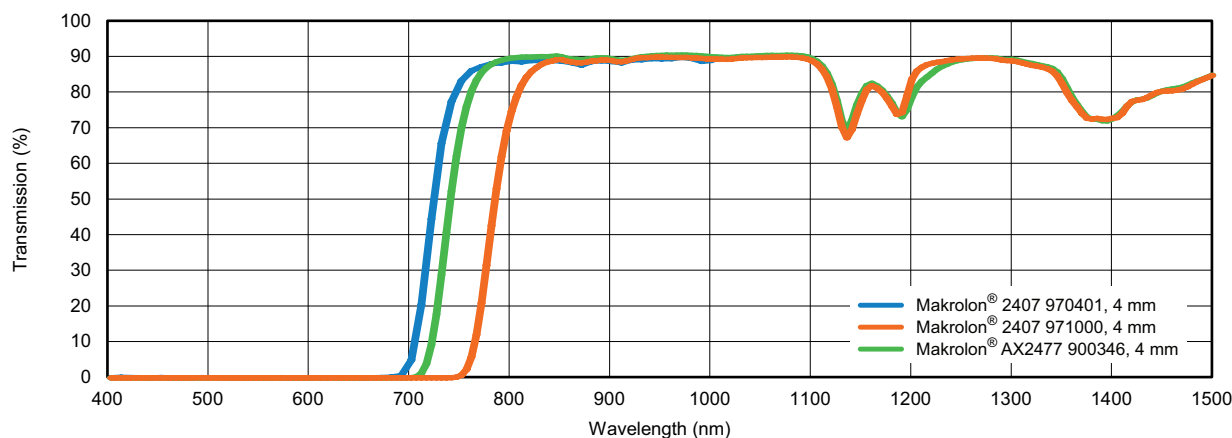


Fig. 2

Customers are recommended to conduct customer's own research and analysis, and decide the suitable materials based on customer's product, operation, and commercial considerations. All trademarks are the property of their respective owners.